

PIO Test Report (-e native)

Verbosity level can be increased via `-v`, `-vv`, or `-vvv` option
Collected 1 tests

Processing test_native in native environment

Building...

Testing...

test/test_native/main.cpp:161: test_proportional_only [PASSED]
test/test_native/main.cpp:162: test_output_is_clamped_high [PASSED]
test/test_native/main.cpp:163: test_output_is_clamped_low [PASSED]
test/test_native/main.cpp:164: test_zero_error_zero_output [PASSED]
test/test_native/main.cpp:165: test_sign_follows_error [PASSED]
test/test_native/main.cpp:166: test_deadband_suppresses_small_error [PASSED]
test/test_native/main.cpp:167: test_deadband_passes_large_error [PASSED]
test/test_native/main.cpp:168: test_deadband_clears_integral [PASSED]
test/test_native/main.cpp:169: test_integral_accumulates [PASSED]
test/test_native/main.cpp:170: test_integral_is_clamped [PASSED]
test/test_native/main.cpp:171: test_integral_unwinds_with_opposite_error [PASSED]
test/test_native/main.cpp:172: test_reset_clears_state [PASSED]
test/test_native/main.cpp:173: test_derivative_reacts_to_change [PASSED]
test/test_native/main.cpp:174: test_derivative_filter_smooths [PASSED]
test/test_native/main.cpp:175: test_set_gains_takes_effect [PASSED]
test/test_native/main.cpp:176: test_set_output_limits [PASSED]
test/test_native/main.cpp:177: test_compute_with_error_matches_compute [PASSED]
test/test_native/main.cpp:178: test_long_stall_does_not_explode_integral [PASSED]
test/test_native/main.cpp:525: test_compass_and_imu_agreeing_is_believed [PASSED]
test/test_native/main.cpp:526: test_a_compass_step_the_imu_does_not_corroborate_is_refused [PASSED]
test/test_native/main.cpp:527: test_a_real_turn_is_not_refused [PASSED]
test/test_native/main.cpp:528: test_a_lasting_disagreement_is_eventually_accepted [PASSED]
test/test_native/main.cpp:529: test_the_first_alignment_is_never_gated [PASSED]
test/test_native/main.cpp:530: test_no_imu_means_no_referee_so_the_compass_is_believed [PASSED]
test/test_native/main.cpp:531: test_gps_speed_the_robot_cannot_reach_is_rejected [PASSED]
test/test_native/main.cpp:532: test_believing_a_bad_gps_speed_pins_the_estimate [PASSED]
test/test_native/main.cpp:533: test_wrap360_positive [PASSED]
test/test_native/main.cpp:534: test_wrap360_negative [PASSED]
test/test_native/main.cpp:535: test_ang_err_wraps_short_way [PASSED]
test/test_native/main.cpp:536: test_wrap_pi_negative [PASSED]
test/test_native/main.cpp:537: test_compass_gives_heading_while_standing_still [PASSED]
test/test_native/main.cpp:538: test_gps_course_wins_while_moving [PASSED]
test/test_native/main.cpp:539: test_falls_back_to_imu_magnetometer_when_compass_dies [PASSED]
test/test_native/main.cpp:540: test_no_absolute_source_is_reported [PASSED]
test/test_native/main.cpp:541: test_health_flags_report_missing_sensors [PASSED]
test/test_native/main.cpp:542: test_a_fix_too_rough_for_speed_is_still_a_working_sensor [PASSED]
test/test_native/main.cpp:543: test_no_position_at_all_is_reported_as_a_missing_gps [PASSED]
test/test_native/main.cpp:544: test_a_rough_fix_is_still_refused_as_a_speed_source [PASSED]
test/test_native/main.cpp:545: test_gyro_carries_heading_when_everything_absolute_is_gone [PASSED]
test/test_native/main.cpp:546: test_imu_dead_uses_gps_course_directly [PASSED]
test/test_native/main.cpp:547: test_imu_and_gps_dead_uses_compass_directly [PASSED]
test/test_native/main.cpp:548: test_mode0_uses_imu_yaw_as_absolute [PASSED]
test/test_native/main.cpp:549: test_mode2_gps_only [PASSED]
test/test_native/main.cpp:550: test_compass_ignored_while_driving_fast [PASSED]
test/test_native/main.cpp:551: test_speed_follows_gps [PASSED]
test/test_native/main.cpp:552: test_speed_falls_back_to_model_without_gps [PASSED]
test/test_native/main.cpp:553: test_open_contactor_does_not_invent_speed [PASSED]
test/test_native/main.cpp:554: test_powered_contactor_still_trusts_the_motor_model [PASSED]
test/test_native/main.cpp:555: test_speed_source_without_imu [PASSED]
test/test_native/main.cpp:556: test_speed_is_clamped [PASSED]
test/test_native/main.cpp:557: test_small_speed_is_squelched [PASSED]
test/test_native/main.cpp:558: test_acceleration_deadband [PASSED]
test/test_native/main.cpp:559: test_wheel_speeds_equal_when_driving_straight [PASSED]
test/test_native/main.cpp:560: test_wheel_speeds_split_when_turning [PASSED]
test/test_native/main.cpp:561: test_rpm_conversion [PASSED]
test/test_native/main.cpp:562: test_odometry_moves_north_when_heading_north [PASSED]
test/test_native/main.cpp:563: test_odometry_moves_east_when_heading_east [PASSED]

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test/test_native/main.cpp:564: test_heading_enu_conversion [PASSED]
test/test_native/main.cpp:565: test_reset_clears_everything [PASSED]
test/test_native/main.cpp:566: test_reset_odom_keeps_heading [PASSED]
test/test_native/main.cpp:567: test_zero_dt_is_ignored [PASSED]
test/test_native/main.cpp:583: test_starts_stopped [PASSED]
test/test_native/main.cpp:584: test_zero_speed_keeps_motors_off [PASSED]
test/test_native/main.cpp:585: test_drives_forward_on_target_heading [PASSED]
test/test_native/main.cpp:586: test_heading_error_steers_right [PASSED]
test/test_native/main.cpp:587: test_heading_error_steers_left [PASSED]
test/test_native/main.cpp:588: test_large_error_spins_on_the_spot [PASSED]
test/test_native/main.cpp:589: test_spin_direction_is_shortest_way [PASSED]
test/test_native/main.cpp:590: test_pwm_never_exceeds_limit [PASSED]
test/test_native/main.cpp:591: test_turn_to_heading_rotates [PASSED]
test/test_native/main.cpp:592: test_turn_to_heading_finishes_inside_tolerance [PASSED]
test/test_native/main.cpp:593: test_turn_to_heading_needs_no_speed_command [PASSED]
test/test_native/main.cpp:594: test_stop_cancels_turn [PASSED]
test/test_native/main.cpp:595: test_manual_forward [PASSED]
test/test_native/main.cpp:596: test_manual_backward [PASSED]
test/test_native/main.cpp:597: test_manual_turn_right [PASSED]
test/test_native/main.cpp:598: test_manual_turn_left [PASSED]
test/test_native/main.cpp:599: test_manual_zero_stops [PASSED]
test/test_native/main.cpp:600: test_manual_straight_latches_the_heading [PASSED]
test/test_native/main.cpp:601: test_hold_waits_for_the_turn_to_settle_before_latching [PASSED]
test/test_native/main.cpp:602: test_hold_does_not_steer_while_it_is_still_settling [PASSED]
test/test_native/main.cpp:603: test_manual_turning_releases_the_hold [PASSED]
test/test_native/main.cpp:604: test_manual_relatches_after_a_turn [PASSED]
test/test_native/main.cpp:605: test_manual_stopped_holds_nothing [PASSED]
test/test_native/main.cpp:606: test_manual_drift_produces_a_correction [PASSED]
test/test_native/main.cpp:607: test_manual_correction_cannot_overpower_the_operator [PASSED]
test/test_native/main.cpp:608: test_manual_speed_loop_runs_on_a_measured_speed [PASSED]
test/test_native/main.cpp:609: test_manual_speed_loop_refuses_to_close_on_the_motor_model [PASSED]
test/test_native/main.cpp:610: test_manual_hold_works_in_reverse [PASSED]
test/test_native/main.cpp:611: test_auto_arcs_below_the_spin_threshold [PASSED]
test/test_native/main.cpp:612: test_auto_spins_above_the_spin_threshold [PASSED]
test/test_native/main.cpp:613: test_spin_threshold_at_180_never_stops_to_turn [PASSED]
test/test_native/main.cpp:614: test_ramp_does_not_jump_to_full_power [PASSED]
test/test_native/main.cpp:615: test_ramp_reaches_target_after_the_ramp_time [PASSED]
test/test_native/main.cpp:616: test_ramp_breakaway_avoids_stiction [PASSED]
test/test_native/main.cpp:617: test_ramp_down_to_zero [PASSED]
test/test_native/main.cpp:618: test_ramp_zero_time_is_instant [PASSED]
test/test_native/main.cpp:619: test_direction_reversal_passes_through_zero [PASSED]
test/test_native/main.cpp:620: test_model_speed_follows_pwm [PASSED]
test/test_native/main.cpp:621: test_reset_clears_outputs [PASSED]
test/test_native/main.cpp:622: test_target_accessors [PASSED]
test/test_native/main.cpp:623: test_setauto_leaves_manual_mode [PASSED]
test/test_native/main.cpp:624: test_auto_steering_keeps_its_sign_in_reverse [PASSED]
test/test_native/main.cpp:625: test_zero_speed_command_stops_an_active_spin [PASSED]
test/test_native/main.cpp:626: test_reset_abandons_a_turn_in_place [PASSED]
test/test_native/main.cpp:627: test_a_turn_that_never_arrives_gives_up [PASSED]
test/test_native/main.cpp:203: test_model_full_duty_no_load [PASSED]
test/test_native/main.cpp:204: test_model_half_duty [PASSED]
test/test_native/main.cpp:205: test_model_current_drops_the_speed [PASSED]
test/test_native/main.cpp:206: test_model_reverse_duty_gives_negative_rpm [PASSED]
test/test_native/main.cpp:207: test_model_mps_conversion [PASSED]
test/test_native/main.cpp:208: test_diode_drop_is_subtracted [PASSED]
test/test_native/main.cpp:209: test_diode_slope_resistance_adds_with_current [PASSED]
test/test_native/main.cpp:210: test_bigger_diode_means_lower_estimate [PASSED]
test/test_native/main.cpp:211: test_two_motors_can_have_different_diodes [PASSED]
test/test_native/main.cpp:212: test_diode_bigger_than_supply_clamps_to_zero [PASSED]
test/test_native/main.cpp:213: test_low_duty_is_not_published [PASSED]
test/test_native/main.cpp:214: test_missing_sensor_is_not_published [PASSED]
test/test_native/main.cpp:215: test_back_emf_never_goes_negative [PASSED]
test/test_native/main.cpp:216: test_stall_detected_after_the_hold_time [PASSED]
test/test_native/main.cpp:217: test_stall_clears_when_the_wheel_turns [PASSED]

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test/test_native/main.cpp:218: test_open_circuit_detected [PASSED]
test/test_native/main.cpp:219: test_open_circuit_not_flagged_during_the_current_ramp [PASSED]
test/test_native/main.cpp:220: test_open_circuit_clear_with_normal_current [PASSED]
test/test_native/main.cpp:221: test_motor_model_reset_clears_state [PASSED]
test/test_native/main.cpp:222: test_config_can_be_changed_at_runtime [PASSED]
test/test_native/main.cpp:799: test_cutoff_scales_with_the_cell_count [PASSED]
test/test_native/main.cpp:800: test_the_same_voltage_is_fine_on_a_smaller_pack [PASSED]
test/test_native/main.cpp:801: test_two_cell_pack_latches_at_its_own_floor [PASSED]
test/test_native/main.cpp:802: test_rails_still_live_after_opening_is_reported [PASSED]
test/test_native/main.cpp:803: test_rails_going_dead_after_opening_is_not_reported [PASSED]
test/test_native/main.cpp:804: test_live_rails_with_the_contactor_closed_are_normal [PASSED]
test/test_native/main.cpp:805: test_safety_input_pulls_oppose_the_active_level [PASSED]
test/test_native/main.cpp:806: test_safety_unwired_button_does_not_estop [PASSED]
test/test_native/main.cpp:807: test_safety_blocks_during_startup_lockout [PASSED]
test/test_native/main.cpp:808: test_safety_allows_motion_when_everything_is_fine [PASSED]
test/test_native/main.cpp:809: test_safety_stops_on_heartbeat_timeout [PASSED]
test/test_native/main.cpp:810: test_safety_stops_on_command_timeout_only_when_moving [PASSED]
test/test_native/main.cpp:811: test_safety_stops_when_agent_is_lost [PASSED]
test/test_native/main.cpp:812: test_safety_software_estop [PASSED]
test/test_native/main.cpp:813: test_safety_hardware_emergency_input [PASSED]
test/test_native/main.cpp:814: test_safety_emergency_input_is_debounced [PASSED]
test/test_native/main.cpp:815: test_safety_absent_button_never_reads_as_pressed [PASSED]
test/test_native/main.cpp:816: test_safety_release_needs_hold_time [PASSED]
test/test_native/main.cpp:817: test_safety_battery_cutoff_needs_to_persist [PASSED]
test/test_native/main.cpp:818: test_relay_is_open_until_the_lockout_expires [PASSED]
test/test_native/main.cpp:819: test_relay_update_lockout_cuts_power_and_holds [PASSED]
test/test_native/main.cpp:820: test_relay_opens_on_emergency_and_battery_cutoff [PASSED]
test/test_native/main.cpp:821: test_relay_stays_open_after_a_flat_pack_even_when_it_recovers [PASSED]
test/test_native/main.cpp:822: test_safety_battery_cutoff_never_unlatches_by_itself [PASSED]
test/test_native/main.cpp:823: test_safety_battery_cutoff_clears_on_restart [PASSED]
test/test_native/main.cpp:824: test_safety_invalid_battery_reading_is_ignored [PASSED]
test/test_native/main.cpp:825: test_safety_rail_voltage_is_not_treated_as_pack [PASSED]
test/test_native/main.cpp:826: test_safety_genuinely_low_pack_still_latches [PASSED]
test/test_native/main.cpp:827: test_safety_battery_tiers [PASSED]
test/test_native/main.cpp:828: test_safety_battery_clear_threshold [PASSED]
test/test_native/main.cpp:829: test_safety_run_state_mapping [PASSED]
test/test_native/main.cpp:830: test_safety_multiple_flags_at_once [PASSED]
test/test_native/main.cpp:832: test_nmea_parses_gga_position [PASSED]
test/test_native/main.cpp:833: test_nmea_rejects_bad_checksum [PASSED]
test/test_native/main.cpp:834: test_nmea_no_fix_is_not_valid [PASSED]
test/test_native/main.cpp:835: test_nmea_parses_rmc_speed_and_course [PASSED]
test/test_native/main.cpp:836: test_nmea_rmc_void_clears_fix [PASSED]
test/test_native/main.cpp:837: test_nmea_southern_and_western_hemisphere [PASSED]
test/test_native/main.cpp:838: test_nmea_ignores_garbage [PASSED]
test/test_native/main.cpp:839: test_nmea_survives_overlong_line [PASSED]
test/test_native/main.cpp:840: test_nmea_freshness_expires [PASSED]
test/test_native/main.cpp:841: test_nmea_gnss_talker_ids_accepted [PASSED]
test/test_native/main.cpp:842: test_nmea_vtg_updates_speed [PASSED]
test/test_native/main.cpp:843: test_nmea_partial_sentence_then_rest [PASSED]
test/test_native/main.cpp:844: test_a_welded_contactor_refuses_motion [PASSED]
test/test_native/main.cpp:845: test_update_lockout_refuses_motion [PASSED]
test/test_native/main.cpp:846: test_soft_battery_stop_does_not_chatter [PASSED]
test/test_native/main.cpp:211: test_a_receiver_with_no_fix_must_not_move_the_speed_estimate [PASSED]
test/test_native/main.cpp:212: test_sim_is_deterministic [PASSED]
test/test_native/main.cpp:213: test_sim_heading_converges_while_driving [PASSED]
test/test_native/main.cpp:214: test_sim_heading_survives_a_disturbed_compass [PASSED]
test/test_native/main.cpp:215: test_sim_speed_tracks_truth [PASSED]
test/test_native/main.cpp:216: test_sim_speed_survives_a_gps_dropout [PASSED]
test/test_native/main.cpp:217: test_sim_dead_reckoning_drift_is_bounded_over_a_short_run [PASSED]
test/test_native/main.cpp:218: test_sim_square_path_returns_near_the_origin [PASSED]
test/test_native/main.cpp:219: test_sim_keeps_navigating_with_each_sensor_lost [PASSED]
test/test_native/main.cpp:195: test_ekf_takes_the_first_fix_without_filtering_it [PASSED]
test/test_native/main.cpp:196: test_ekf_uncertainty_grows_while_coasting_and_shrinks_on_a_fix [PASSED]
test/test_native/main.cpp:197: test_ekf_trusts_rtk_far_more_than_plain_gps [PASSED]

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test/test_native/main.cpp:198: test_ekf_rejects_a_multipath_jump [PASSED]
test/test_native/main.cpp:199: test_ekf_gives_in_if_the_receiver_really_has_moved [PASSED]
test/test_native/main.cpp:200: test_ekf_beats_raw_gps_and_dead_reckoning [PASSED]
test/test_native/main.cpp:201: test_ekf_coasts_through_a_gps_dropout [PASSED]
test/test_native/main.cpp:107: test_zero_offset_changes_nothing [PASSED]
test/test_native/main.cpp:108: test_antenna_ahead_while_facing_north [PASSED]
test/test_native/main.cpp:109: test_antenna_ahead_while_facing_east [PASSED]
test/test_native/main.cpp:110: test_antenna_to_the_left_while_facing_north [PASSED]
test/test_native/main.cpp:111: test_spinning_on_the_spot_does_not_move_the_centre [PASSED]
test/test_native/main.cpp:112: test_offset_diagonal_round_trips_at_every_heading [PASSED]
test/test_native/main.cpp:113: test_tangential_speed_of_a_spinning_sensor [PASSED]
test/test_native/main.cpp:153: test_defaults_come_from_the_table [PASSED]
test/test_native/main.cpp:154: test_an_unknown_key_is_refused_not_created [PASSED]
test/test_native/main.cpp:155: test_a_value_in_range_is_kept_and_marked_overridden [PASSED]
test/test_native/main.cpp:156: test_out_of_range_is_refused_and_changes_nothing [PASSED]
test/test_native/main.cpp:157: test_the_bounds_themselves_are_allowed [PASSED]
test/test_native/main.cpp:158: test_nan_is_refused_before_the_range_check [PASSED]
test/test_native/main.cpp:159: test_stopped_only_is_refused_while_moving [PASSED]
test/test_native/main.cpp:160: test_live_parameters_apply_while_moving [PASSED]
test/test_native/main.cpp:161: test_reset_restores_the_default_and_clears_the_override [PASSED]
test/test_native/main.cpp:162: test_reset_of_a_stopped_parameter_also_needs_a_stop [PASSED]
test/test_native/main.cpp:163: test_reset_all_leaves_stopped_parameters_alone_while_moving [PASSED]
test/test_native/main.cpp:164: test_storage_keys_are_short_enough_and_distinct [PASSED]
test/test_native/main.cpp:144: test_seeds_from_the_header_when_storage_is_empty [PASSED]
test/test_native/main.cpp:145: test_storage_beats_the_header_on_every_later_boot [PASSED]
test/test_native/main.cpp:146: test_setting_a_known_ssid_replaces_rather_than_duplicates [PASSED]
test/test_native/main.cpp:147: test_order_is_priority_and_can_be_changed [PASSED]
test/test_native/main.cpp:148: test_removing_keeps_the_rest_in_order [PASSED]
test/test_native/main.cpp:149: test_the_list_is_bounded [PASSED]
test/test_native/main.cpp:150: test_an_over_long_ssid_is_truncated_not_overflowed [PASSED]
test/test_native/main.cpp:151: test_an_empty_ssid_is_refused [PASSED]
test/test_native/main.cpp:296: test_help_lists_the_commands [PASSED]
test/test_native/main.cpp:297: test_unknown_command_says_so [PASSED]
test/test_native/main.cpp:298: test_params_lists_every_parameter [PASSED]
test/test_native/main.cpp:299: test_get_and_set_round_trip [PASSED]
test/test_native/main.cpp:300: test_out_of_range_is_refused_with_a_reason [PASSED]
test/test_native/main.cpp:301: test_a_word_is_not_a_number [PASSED]
test/test_native/main.cpp:302: test_unknown_key_is_named [PASSED]
test/test_native/main.cpp:303: test_reset_restores_the_default [PASSED]
test/test_native/main.cpp:304: test_wifi_list_never_shows_a_password [PASSED]
test/test_native/main.cpp:305: test_wifi_password_may_contain_spaces [PASSED]
test/test_native/main.cpp:306: test_wifi_add_delete_and_reorder [PASSED]
test/test_native/main.cpp:307: test_stopped_only_parameter_is_refused_while_moving [PASSED]
test/test_native/main.cpp:308: test_an_overlong_line_is_dropped_not_truncated [PASSED]
test/test_native/main.cpp:309: test_commands_split_across_reads_still_work [PASSED]
test/test_native/main.cpp:310: test_blank_lines_are_ignored [PASSED]
test/test_native/main.cpp:311: test_agent_can_be_set_by_hand [PASSED]
test/test_native/main.cpp:312: test_agent_accepts_a_hostname_too [PASSED]
test/test_native/main.cpp:313: test_agent_auto_returns_to_discovery [PASSED]
test/test_native/main.cpp:314: test_agent_with_no_argument_reports_where_it_is_looking [PASSED]
test/test_native/main.cpp:315: test_agent_says_so_when_it_cannot_store [PASSED]
test/test_native/main.cpp:277: test_a_perfect_field_sweep_without_turning_is_refused [PASSED]
test/test_native/main.cpp:278: test_a_real_turn_with_the_gyro_agreeing_passes [PASSED]
test/test_native/main.cpp:279: test_rocking_back_and_forth_does_not_count [PASSED]
test/test_native/main.cpp:280: test_without_an_imu_the_field_measure_still_works [PASSED]
test/test_native/main.cpp:281: test_yaw_only_calibration_gives_a_usable_reference [PASSED]
test/test_native/main.cpp:282: test_unswept_z_is_left_alone [PASSED]
test/test_native/main.cpp:283: test_a_real_three_axis_sweep_still_scales_z [PASSED]
test/test_native/main.cpp:284: test_standing_still_makes_no_progress [PASSED]
test/test_native/main.cpp:285: test_a_full_turn_covers_the_whole_circle [PASSED]
test/test_native/main.cpp:286: test_half_a_turn_is_refused [PASSED]
test/test_native/main.cpp:287: test_a_noisy_nudge_is_not_a_turn [PASSED]
test/test_native/main.cpp:288: test_clear_calibration_returns_to_raw [PASSED]
test/test_native/main.cpp:289: test_restarting_forgets_the_previous_sweep [PASSED]

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test/test_native/main.cpp:122: test_zero_angles_are_the_identity [PASSED]
test/test_native/main.cpp:123: test_angles_round_trip [PASSED]
test/test_native/main.cpp:124: test_yaw_ninety_turns_forward_into_left [PASSED]
test/test_native/main.cpp:125: test_roll_ninety_puts_up_on_its_side [PASSED]
test/test_native/main.cpp:126: test_an_empty_quaternion_is_treated_as_no_rotation [PASSED]
test/test_native/main.cpp:127: test_a_slightly_off_unit_quaternion_is_repaired [PASSED]
test/test_native/main.cpp:128: test_the_mode_picks_one_form_and_ignores_the_other [PASSED]
test/test_native/main.cpp:129: test_pitch_at_the_limit_does_not_produce_nan [PASSED]
test/test_native/main.cpp:108: test_imu_rates_survive_millis_rollover [PASSED]
test/test_native/main.cpp:109: test_imu_window_restarts_across_rollover_and_forgets_old_traffic [PASSED]
test/test_native/main.cpp:110: test_imu_window_expires_without_events_or_diagnostic_reads [PASSED]
test/test_native/main.cpp:111: test_imu_manual_reset_restarts_counts_and_warmup [PASSED]
----- native:test_native [PASSED] Took 0.21 seconds -----
```

```
===== SUMMARY =====
Environment      Test      Status      Duration
-----
native          test_native PASSED      00:00:00.214
===== 261 test cases: 261 succeeded in 00:00:00.214 =====
```